1. Student Marks Analysis by using(List)

Problem Statement

Write a Python program to store marks of students in a **list** and display the **maximum marks, minimum marks, average marks, and sorted list**.

Explanation

The program stores student marks in a list and uses built-in functions to analyze the data.

Input

List of student marks.

Example:

[78, 65, 89, 90, 56]

Output

Maximum Marks: 90
Minimum Marks: 56
Average Marks: 75.6
Sorted Marks: [56, 65, 78, 89, 90]

Code:-

```
marks = [78, 65, 89, 90, 56]#createing list ; list:it is like storing numbers in box

print("Marks:", marks)

print("Maximum Marks:", max(marks))#  max() finds the biggest number in the list.

print("Minimum Marks:", min(marks))#  min() finds the smallest number.

print("Average Marks:", sum(marks)/len(marks))#sum(marks)- Adds all numbers, len(marks)- Counts how many numbers are there
print("Sorted Marks:", sorted(marks))#sorted() arranges numbers from small to big.
```

Test Cases

Input	Output
[10,20,30,40]	Max:40 Min:10 Avg:25 Sorted:[10,20,30,40]
[50,60,70]	Max:70 Min:50 Avg:60 Sorted:[50,60,70]

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2. Student Database by using(Dictionary)

Problem Statement

Create a program to store **student names and marks using a dictionary** and display the student with the highest marks.

Explanation

Dictionary keys represent student names and values represent marks.

Input

Rahul 85
Anita 92
Kiran 76

Output

Student with highest marks: Anita
Marks: 92

Code:-

```
students = {
    "Rahul": 85,
    "Anita": 92,
    "Kiran": 76
}
```

A dictionary stores **keys and value pairs**.

print("Student Marks:", students)# Shows all student marks.

highest = max(students, key=students.get)

print("Student with highest marks:", highest)# This finds the **student with highest marks**.

print("Marks:", students[highest])

Test Cases

Input	Output
{"A":50,"B":80,"C":60}	B with 80
{"Tom":70,"Sam":90}	Sam with 90

---

3. Unique Word Extractorby using (Set)

Problem Statement

Write a Python program that accepts a sentence and prints all **unique words using a set**.

Explanation

The program splits the sentence into words and removes duplicates using a set.

Input

python is easy and python is powerful

Output

{'python','is','easy','and','powerful'}

Code:-

s = "python is easy and python is powerful"

We store a sentence in a variable.

words = s.lower().split()

unique_words = set(words)# **set removes duplicates**

print("Unique Words:", unique_words)

Test Cases

Input	Output
hello hello world	{'hello','world'}
python python code	{'python','code'}

---

4. Product Inventory by using(Tuple + Dictionary)

Problem Statement

Create a program to store product information using **dictionary where value is a tuple (price, quantity)**.

Explanation

Each product has a price and quantity stored in a tuple.

Input

Laptop (50000,10)
Mouse (500,50)
Keyboard (1200,20)

Output

Laptop → Price:50000 Quantity:10
Mouse → Price:500 Quantity:50
Keyboard → Price:1200 Quantity:20

Code:-

```
products = {
    "Laptop": (50000, 10),
    "Mouse": (500, 50),
    "Keyboard": (1200, 20)
}
# Dictionary stores products.
for product, details in products.items():    # This loop checks each product one by one.
  price, quantity = details # This separates the tuple.
    print(product, "→ Price:", price, "Quantity:", quantity)
```

Test Cases

Input	Output
Pen (10,100)	Price:10 Quantity:100
Book (50,30)	Price:50 Quantity:30

---

5. Remove Duplicate Numbers by using (List + Set)

Problem Statement

Write a Python program to remove duplicate numbers from a list.

Explanation

Convert the list into a set to remove duplicates.

Input

[1,2,3,2,4,5,1]

Output

[1,2,3,4,5]

Code:-

```
numbers = [1,2,3,2,4,5,1]

# createing list

unique_numbers = list(set(numbers))

#converting list-set and Convert set back to list.

print("Original List:", numbers)

print("List after removing duplicates:", unique_numbers)
```

Test Cases

Input	Output
[1,1,1,2]	[1,2]
[5,5,6,7]	[5,6,7]

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6. Temperature Analysis by using (Array)

Problem Statement

Store weekly temperatures using the **array module** and calculate maximum, minimum, and average temperature.

Explanation

The array module is used to store numeric values efficiently.

Input

[30,32,31,29,35,34,33]

Output

Highest Temperature: 35
Lowest Temperature: 29
Average Temperature: 32

Code:-

```
from array import *


temps = array('i',[30,32,31,29,35,34,33])
```

createing an **array of temperatures**.from Sunday to saturday

```
print("Temperatures:", temps)


print("Highest Temperature:", max(temps))#finding highest temp
print("Lowest Temperature:", min(temps))#findig lowest temp
print("Average Temperature:", sum(temps)/len(temps))# Finding average temperature.
```

Test Cases

Input	Output
[20,22,21]	Max:22 Min:20 Avg:21
[10,15,20]	Max:20 Min:10 Avg:15

---

7. Contact Book by using (Dictionary)

Problem Statement

Create a contact management program using a dictionary.

Explanation

Names are keys and phone numbers are values.

Input

Dhanush:9876543210
Ravi:9123456789

Output

Contacts:
Dhanush → 9876543210
Ravi → 9123456789

Code:-

```
contacts = {
    "sara": "9876543210",
    "tara": "9123456789"
}


print("Contacts:")


for name, number in contacts.items():
    print(name, "→", number)
```

Test Cases

Input	Output
A:111	A →111
B:222	B →222

---

8. Word Frequency Counter by using (Dictionary)

Problem Statement

Write a program to count the number of occurrences of each word in a sentence.

Explanation

Use a dictionary where words are keys and counts are values.

Input

python is easy and python is powerful

Output

python:2
is:2
easy:1
and:1
powerful:1

code:-

```
sentence = "python is easy and python is powerful"

# python counts how many times each word appears.

words = sentence.split()

frequency = {}


for word in words:

    frequency[word] = frequency.get(word, 0) + 1


for word, count in frequency.items():

    print(word, ":", count)
```

Test Cases

Input	Output
hello hello world	hello:2 world:1
a b a b c	a:2 b:2 c:1

---

9. Shopping Cart Total by using(List + Tuple)

Problem Statement

Create a shopping cart system where items are stored as tuples (item_name, price) inside a list.

Explanation

The program calculates total price and displays all items.

Input

[("pen",10),("book",50),("bag",500)]

Output

Items in Cart:
pen - 10
book - 50
bag - 500

Total Cost: 560

Code:-

```
sentence = "python is easy and python is powerful"


words = sentence.split()

frequency = {}


for word in words:

    frequency[word] = frequency.get(word, 0) + 1


for word, count in frequency.items():

    print(word, ":", count)
```

Test Cases

Input	Output
[("apple",20),("banana",10)]	Total:30
[("shirt",500),("cap",200)]	Total:700